

■ UNIT-I

Q1. With a neat diagram, explain the different components of the CPU (8 Marks)

How to write in exam:

- Definition of CPU as the brain of computer
- Explain instruction cycle (fetch–decode–execute)
- ALU
 - Arithmetic operations
 - Logical operations
 - Forensic link (hashing, bitwise ops)
- Control Unit
 - Fetches instructions
 - Decodes & generates control signals
 - Coordinates CPU components
- Registers
 - PC, IR, MAR, MBR, ACC
 - Temporary high-speed storage
- Cache Memory
 - L1, L2, L3
 - Reduces memory access time
- System Bus
 - Data bus
 - Address bus
 - Control bus

- Draw neat labelled diagram
 - Forensic importance
 - Process execution
 - Memory analysis
 - Conclusion
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Q2. Explain Process Control Block (PCB) and the information it maintains (8 Marks)

- Definition of PCB
- Role in multitasking & context switching
- Process identification
 - PID, PPID, UID
- Processor state
 - PC, registers, stack pointer
- Process state
 - New, Ready, Running, Waiting, Terminated
- Memory management info
 - Page tables, segment tables
- Accounting information
 - CPU time, priority
- I/O status
 - Open files, devices
- Forensic relevance
 - Identify malicious or suspended processes
- Conclusion

Q3. Explain Volume Boot Record (VBR) and GPT and the data they hold (8 Marks)

Volume Boot Record (VBR)

- First sector of a partition
- Bootstrapping code
- BIOS Parameter Block (BPB)
 - Bytes per sector
 - Sectors per cluster
 - FAT count
- NTFS VBR stores MFT location

GPT (GUID Partition Table)

- Introduced with UEFI
- Protective MBR
- Primary GPT Header
 - Disk GUID
 - CRC checksum
- Partition Entry Array
 - Partition GUIDs
 - Start & end LBA
- Backup GPT Header

Forensic importance

- Partition recovery
- Disk integrity validation
- Anti-forensics detection

- Conclusion
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Q4. Convert 450 (decimal) to hexadecimal and 456 (decimal) to binary (8 Marks)

Decimal → Hexadecimal (450)

- Explain division by 16 method
- Show all steps clearly
- Remainders → bottom-to-top
- Final answer: $1C2_{16}$

Decimal → Binary (456)

- Explain division by 2 method
- Stepwise quotient & remainder table
- Final binary value

Forensic importance

- Hex & binary used in:
 - Disk sectors
 - Memory dumps
 - File headers
 - Conclusion
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Q7. Explain Locard's Principle of Exchange (Digital Forensics) (8 Marks)

- Define Locard's principle
- Digital interpretation
- File system traces

- MAC times
 - Journal logs
 - Registry traces
 - RecentDocs
 - USBSTOR
 - Log files
 - Login, application, system logs
 - Network artifacts
 - IP, DNS, proxy logs
 - Memory artifacts
 - RAM remnants
 - Forensic importance
 - Timeline reconstruction
 - Attribution
 - Anti-forensics detection
 - Conclusion
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Q8. Explain Master Boot Record (MBR) with neat diagram (8 Marks)

- Definition of MBR
- Location: LBA 0
- Size: 512 bytes
- Structure
 - Bootloader code (446 bytes)
 - Partition table (64 bytes)

- Boot signature (55AA)
 - Partition table entries explanation
 - Limitations:
 - 2 TB disk size
 - 4 primary partitions
 - Forensic importance
 - Bootkit detection
 - Hidden partition recovery
 - Draw diagram
 - Conclusion
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Q9. Describe HDD and SSD internal structures (8 Marks)

HDD Structure

- Platters
- Spindle motor
- Read/write heads
- Actuator arm
- Tracks, sectors, clusters
- Forensic behavior:
 - Easier recovery
 - Fragmentation

SSD Structure

- NAND flash cells
- Pages & blocks

- Flash Translation Layer (FTL)
- Controller & DRAM cache
- Forensic behavior:
 - TRIM
 - Wear leveling
 - Harder recovery

Comparison + forensic impact

- Conclusion
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Q10. Explain OS Boot Process in detail (8 Marks)

- POST
 - BIOS/UEFI initialization
 - Bootloader loading
 - Stage 1 & Stage 2 loaders
 - Kernel loading
 - Driver & hardware initialization
 - System services start
 - User login
 - Forensic importance
 - Bootkits
 - Persistence mechanisms
 - Conclusion
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Q11. Explain Process Management in an Operating System (8 Marks)

- Definition of process
 - Process states
 - Schedulers
 - Long-term
 - Short-term
 - Medium-term
 - Scheduling algorithms
 - FCFS
 - SJF
 - Round Robin
 - Priority
 - Context switching
 - IPC mechanisms
 - Threads & multithreading
 - Forensic relevance
 - Malware behavior
 - Parent-child process analysis
 - Conclusion
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Q12. Explain Memory Management in an Operating System (8 Marks)

- Definition of memory management
- Memory allocation techniques
- Paging

- Segmentation
 - Virtual memory
 - Demand paging & page faults
 - Memory protection
 - Data structures:
 - Page tables
 - TLB
 - Forensic importance
 - RAM artifacts
 - Encryption keys
 - Malware payloads
 - Conclusion
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UNIT-II

Q1. Define Digital Forensics and explain the Digital Forensics Investigation Process

How to write (8 marks):

- Definition of Digital Forensics (scientific + legal process)
- Scope: computers, mobiles, networks, cloud, IoT
- Importance of legality and court admissibility

Digital Forensics Investigation Process

1. Identification

- Identify devices, data sources, evidence types

- Examples: emails, logs, deleted files
- 2. Preservation
 - Prevent modification
 - Write-blockers, hashing, Faraday bags
- 3. Collection / Acquisition
 - Bit-by-bit, logical, live, targeted
 - Forensic tools (FTK Imager, EnCase)
- 4. Examination
 - Keyword search, deleted file recovery
 - File system, registry, log analysis
- 5. Analysis
 - Who, what, when, how, why
 - Correlation of technical evidence
- 6. Presentation
 - Reports, timelines, exhibits
 - Explanation for non-technical audience

Conclusion

- Structured process ensures integrity and admissibility
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Q2. Explain Locard's Principle of Exchange and its relevance in Digital Forensics

- Statement of Locard's Principle ("Every contact leaves a trace")
- Extension from physical to digital domain

Digital traces created through:

1. File system artifacts

- MAC timestamps, journals, metadata
- 2. Registry traces
 - RecentDocs, Run keys, USBSTOR, MRU
- 3. Log files
 - Authentication, application, system logs
- 4. Network artifacts
 - DNS cache, firewall, proxy, IP logs
- 5. Memory artifacts
 - Running processes, keys, malware code

Relevance in Digital Forensics

- Establishes user activity
- Incident reconstruction & timeline
- Attribution to user/system
- Detection of anti-forensics

Conclusion

- Foundation principle of all digital investigations
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Q3. Explain handling of a Digital Crime Scene with proper documentation steps

- Importance of procedural accuracy and legality

Steps:

1. Securing and isolating the scene
 - Restrict access, disable networks
2. Initial observation
 - Identify devices, screens, peripherals

3. Documentation

- Photos, serial numbers, timestamps

4. Volatile data collection

- RAM, processes, network connections

5. Proper shutdown & seizure

- Controlled shutdown after volatile capture

6. Packaging & transportation

- Anti-static and Faraday bags

7. Chain-of-Custody

- Record every transfer

8. Legal considerations

- Warrants, scope, privacy laws

Conclusion

- Proper handling ensures evidence reliability
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Q4. Explain different types of Electronic Evidence and important documents used

Types of Electronic Evidence

1. Computer evidence

- HDD/SSD, RAM, logs, documents

2. Mobile evidence

- Calls, messages, app data, GPS

3. Network evidence

- Firewall, router, IDS logs

4. Cloud evidence

- Login records, access logs
- 5. IoT evidence
 - Smart devices, vehicle data
- 6. Removable media
 - USBs, memory cards

Important Documents

- Search warrant / legal authorization
- Chain-of-custody form
- Evidence collection form
- Investigator's notes
- Consent form
- Case summary report

Conclusion

- Documentation ensures admissibility
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Q5. Explain Chain-of-Custody and its importance

- Definition of Chain-of-Custody (chronological handling record)

Elements

- Case information
- Evidence description
- Transfer log
- Seal information

Steps

- Identification & labeling

- Secure storage
- Authorized access only
- Continuous documentation

Importance

- Maintains evidence integrity
- Prevents tampering
- Mandatory for court
- Builds trust in investigation
- Supports repeatability

Consequences of failure

- Evidence rejection
- Case dismissal

Conclusion

- Backbone of digital forensics
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Q6. Explain Write-Blockers and their purpose

- Definition of write-blocker
- Read-only access to evidence

Purpose

- Preserve original evidence
- Maintain legal admissibility
- Prevent contamination
- Enable safe imaging

Types

1. Hardware write-blockers
 - Most reliable, lab standard
2. Software write-blockers
 - OS-level, less secure

Working

- Blocks write/delete/format commands

Importance

- Prevents timestamp modification
- Required by NIST/SWGDE

Conclusion

- Essential forensic safeguard
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Q7. Explain Imaging Techniques used in Digital Forensics

- Definition of forensic imaging

Techniques

1. Bit-stream imaging
 - Sector-by-sector copy (gold standard)
2. Logical imaging
 - Selected files only
3. Live imaging
 - Powered-on systems, encrypted drives
4. Sparse imaging
 - Only allocated clusters
5. Targeted imaging

- Specific artifacts

6. Remote imaging

- Network/cloud environments

Validation

- Hashing before and after imaging

Conclusion

- Choice depends on case & constraints
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Q8. Explain Evidence Integrity and role of hashing

- Definition of evidence integrity
- Fragile nature of digital evidence

Hashing

- Definition of hash function
- MD5, SHA-1, SHA-256

During acquisition

- Hash original
- Hash image
- Compare values

During analysis

- Verify transfers
- Malware identification
- Known-file filtering

Importance

- Detects alteration

- Ensures reproducibility
- Legal defensibility

Conclusion

- Hashing = digital fingerprint
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Q9. Differentiate Data Recovery and Data Carving

Data Recovery

- Uses file system metadata
- Recovers filenames & timestamps
- Depends on intact structures

Data Carving

- Uses file signatures
- Works without metadata
- Effective on corrupted disks

Comparison

- Accuracy
- Speed
- Use cases

Forensic usage

- Insider threats → recovery
- Destroyed metadata → carving

Conclusion

- Complementary techniques
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Q10. Explain the Acquisition Process in Digital Forensics

Steps

1. Preparation
2. Device documentation
3. Use of write-blocker
4. Selection of acquisition method
5. Hashing before acquisition
6. Imaging process
7. Hashing after acquisition
8. Documentation & secure storage

Conclusion

- Acquisition is foundation of forensic analysis
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Q11. Explain SOPs for acquiring and preserving digital evidence

- Definition of SOPs

Key SOP areas

- Planning & authorization
- Scene security
- Volatile data capture
- Safe shutdown
- Write-blocker usage
- Forensic imaging
- Preservation & storage
- Chain-of-custody

- Reporting

Conclusion

- SOPs ensure consistency & legality
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Q12. Importance of Documentation and Preservation of Evidence

Documentation importance

- Transparency
- Reproducibility
- Chain-of-custody
- Court presentation

Preservation importance

- Prevents modification
- Long-term availability
- Environmental protection
- Legal admissibility

Techniques

- Write-blockers
- Faraday bags
- Anti-static storage
- Secure lockers

Conclusion

- Without documentation & preservation, evidence fails
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UNIT-III (WINDOWS FORENSICS)

Q1. Explain Windows Log Sources and their Forensic Importance (8 Marks)

Write structure:

- Definition of Windows log sources as system-generated records
- Purpose of logs as digital footprints
- Windows Event Logs
 - .evt / .evtx format
 - Storage location
- Security Logs
 - Login success/failure
 - Privilege escalation
- System Logs
 - Boot/shutdown

- Driver loading
 - Application Logs
 - Antivirus, browser, email logs
 - Registry-based logs
 - USB usage
 - Program execution history
 - Forensic importance
 - Timeline reconstruction
 - Intrusion detection
 - Court-admissible evidence
 - Conclusion
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Q2. Explain Windows Registry Files and their Forensic Significance (8 Marks)

- Define Windows Registry
- Explain hierarchical structure (hives, keys, values)
- SYSTEM hive
 - Hardware, drivers, boot config
- SOFTWARE hive
 - Installed programs
- SAM
 - User accounts, password hashes
- SECURITY
 - Audit and security policies
- NTUSER.DAT

- User activity and preferences
 - Registry file locations
 - Forensic significance
 - User activity tracking
 - Malware persistence detection
 - Timeline analysis
 - Conclusion
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Q3. Explain Windows Events and Event IDs in Forensic Analysis (8 Marks)

- Define Windows events
- Meaning of Event IDs
- Event storage in Event Logs
- Event types
 - Information
 - Warning
 - Error
 - Critical
- Common forensic event categories
 - Logon/logoff
 - Account management
 - System boot/shutdown
 - Application execution
- Forensic importance
 - Unauthorized access detection

- Incident time identification
 - User attribution
 - Conclusion
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Q4. Explain Different Types of Windows Logs (8 Marks)

- Define Windows logs
- System Logs
 - OS operations
 - Hardware & driver issues
- Security Logs
 - Authentication
 - Privilege usage
- Application Logs
 - Software errors
 - Malware behavior
- Setup Logs
 - OS install & updates
- Forwarded Events
 - Centralized enterprise logging
- Log storage location
- Forensic importance
 - Misuse detection
 - Timeline creation
- Conclusion

Q5. Explain FAT File System and its Forensic Importance (8 Marks)

- Definition of FAT file system
- Cross-platform usage (USB, SD cards)
- Types
 - FAT12
 - FAT16
 - FAT32
- Common FAT structure
 - Boot sector (VBR)
 - FAT table
 - Root directory
 - Data area
- File deletion behavior (0xE5)
- Slack space concept
- Forensic importance
 - Easy deleted file recovery
 - Common in data theft cases
- Conclusion

Q6. Explain NTFS File System and its Forensic Significance (8 Marks)

- Definition of NTFS
- NTFS structure
 - Boot sector

- Master File Table (MFT)
 - System files (\$LogFile, \$Bitmap, \$MFTMirr)
 - Data area
 - Role of MFT
 - Journaling feature
 - Multiple timestamps (MAC)
 - Forensic significance
 - Deleted file recovery
 - Timeline reconstruction
 - Anti-forensics detection
 - Conclusion
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UNIT-IV (LINUX FORENSICS)

Q1. Explain Linux Log Sources and their Forensic Importance (8 Marks)

- Define Linux log sources
- Default log directory (/var/log)
- syslog / messages
 - General system activity
- auth.log / secure
 - Login, SSH, sudo
- kern.log
 - Kernel & driver events
- dmesg

- Hardware & USB connections
 - boot.log
 - Startup processes
 - Binary logs
 - wtmp, btmp, lastlog
 - Forensic importance
 - Intrusion detection
 - Privilege escalation analysis
 - Conclusion
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Q2. Explain Recreation of EXT Partitions in Linux Forensics (8 Marks)

- Define EXT partition recreation
- EXT variants (EXT2, EXT3, EXT4)
- Important EXT components
 - Superblock
 - Block groups
 - Inode tables
 - Data blocks
 - Journal
- Recreation steps
 - Identify EXT signatures
 - Superblock analysis
 - Block group & inode reconstruction
 - Journal analysis

- Rebuild partition table
 - Tools used (TestDisk, Sleuth Kit)
 - Forensic importance
 - Recover deleted partitions
 - Detect anti-forensics
 - Conclusion
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UNIT-V (macOS FORENSICS)

Q1. Explain Log Types of macOS and their Forensic Importance (8 Marks)

- Define macOS logging system
- UNIX-based logging architecture
- System Logs
 - Startup, kernel, hardware
- Security Logs
 - Login, sudo, authentication
- Application Logs
 - Safari, Chrome, Mail
- Install Logs
 - Software & update installation
- Unified Logging System
 - log show, log collect
- Forensic importance
 - Timeline reconstruction

- Intrusion detection
 - Conclusion
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Q2. Explain Basics of HFS and HFS+ File Systems with Forensic Importance (8 Marks)

HFS

- Legacy Apple file system
- B-tree based structure
- Components:
 - Volume header
 - Catalog file
 - Extent overflow file
- No journaling
- Forensic relevance (legacy recovery)

HFS+

- Improved HFS
 - Unicode filenames
 - Journaling support
 - Metadata-rich structure
 - Forensic importance
 - Deleted file recovery
 - Timeline analysis
 - Conclusion
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Q3. Explain Basics of APFS File System and its Forensic Importance (8 Marks)

- Define APFS
 - Designed for SSDs
 - Key features
 - Containers & volumes
 - Copy-on-Write
 - Snapshots
 - Encryption
 - APFS components
 - Container
 - Volumes
 - Object map
 - Snapshots
 - Forensic importance
 - Snapshot-based recovery
 - Metadata protection
 - Challenges
 - Encryption
 - CoW complicates carving
 - Conclusion
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✅ FINAL EXAM TIP (VERY IMPORTANT)

For EVERY 8-MARK ANSWER, follow this fixed pattern:

1. Definition – 4 lines
2. Core explanation – 6 to 8 bullet points

3. Forensic importance – separate heading

4. 2-line conclusion

→ This alone is enough to score 7–8 marks consistently.
